# ANKITHA SUDARSHAN

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## EDUCATION

**University at Buffalo, SUNY** • *PhD in Computer Science* Aug 2024 – 2028 (expected)
Advisor: Dr. Rohini Srihari • Focus: Multimodal AI, 3D Computer Vision, Fine Grained Human Motion Evaluation

**Purdue University** • *MS in Computer Science* Aug 2022 – May 2024

**Columbia University** • *Machine Learning Summer School (MLSS NYC 2026)* • In collaboration with Bloomberg 2026
**Selected as one of ~200 PhD researchers worldwide to participate and present research; competitively admitted from a global applicant pool**.

## PUBLICATIONS & PATENTS

- "Transition Correctness: A Reference-Based Measure for Path and Timing Fidelity in Fine-Grained Motion." *Under review, ICLR 2027.*
- "Investigating Fine-Grained Structure in Discrete 3D Human Motion Representations." *Under review, 3DV 2027.*
- "Low-Resource Rhythm Learning of South Asian Beat Structures." *PMLR Vol. 303, 2026 • AAAI EAAI 2026 • NeurIPS WiML 2025.*
- "Footwork2Framework: A Large-Scale Dataset and Generative Framework for Expressive, Fine-Grained Audio-Driven 3D Human Motion." *Under review, WACV 2027.*
- "Improving Contextual Recognition in ASR via Semantic Lattice Rescoring." *arXiv, 2023.*
- ***Patent***: "System and Method for Image Analysis and Summarization" (2020) • IEDC Cell, Govt. of India, Dept. of Science & Technology

## RESEARCH EXPERIENCE

**Transition Correctness: A Reference-Based Measure for Path and Timing Fidelity in Fine-Grained Motion** | *Under review, ICLR 2027* Aug 2026 – Present
*University at Buffalo*
- Designed a three-stage reference-based measure for motion transition correctness operating in a unified joint-angle (DOF) space: (1) a physics-based gate checking anatomical feasibility per frame, body-part-wise; (2) path correctness scoring the trajectory through angle-space against a reference corridor built from expert clips; (3) timing correctness reusing the same angle-space data -- decomposing deviation into separable spatial and pacing components that DTW-based measures conflate.
- Validated via controlled perturbation experiments on SMPL-X motion (Bharatanatyam), surgical kinematics (JIGSAWS), and sign language (SignAvatars); applicable to generative motion quality assessment, scene interaction evaluation, and synthetic data validation; requires only 2-3 expert reference demonstrations, no error-labeled training data.

**Investigating Fine-Grained Structure in Discrete 3D Human Motion Representations** | *Under review, 3DV 2027* May 2026 – July 2026
*University at Buffalo*
- Trained VQ-VAE on 3D body + articulated hand motion without category or performer labels; probed codebook assignments across 18 motion classes -- **body representations achieved 58.1% recognition accuracy vs. 22.6% majority-class baseline**, demonstrating reconstruction-only objectives implicitly encode category-relevant structure.
- Body+hand representations reached 53.4% movement-category accuracy; hand-only features retained stronger performer-specific information -- showing body and hand motion encode fundamentally different performance aspects, with codebook tokens aligning to visually consistent poses across performers.

**Footwork2Framework: A Large-Scale Dataset and Generative Framework for Expressive, Fine-Grained Audio-Driven 3D Human Motion** | *Under review, WACV 2027* May 2025 – August 2025
*University at Buffalo*
- Released the first 3D motion dataset with synchronized full-body + hand mesh annotations (1,910 clips, 47,750s -- 2.5× larger than AIST++); developing VQ-VAE generation framework for 3D human motion modeling conditioned on motion and audio, with bidirectional audio-motion alignment and appearance retargeting for human avatar synthesis.

**Low-Resource Rhythm Learning of South Asian Beat Structures** | *PMLR 2026 • AAAI EAAI 2026 • NeurIPS WiML 2025* Sep – Nov 2025
*University at Buffalo*
- Constructed first annotated structured beat dataset (31,560s, 7 classes); **hybrid MFCC+CLAP model achieved 76.2% accuracy (F1 0.73) vs. 69.8% baseline (F1 0.66)**; zero-shot transfer to rare classes under 30:1 imbalance (recall 0.71, 0.56).

**Improving Contextual Recognition in ASR via Semantic Lattice Rescoring** | *arXiv 2023 • Purdue* Oct 2022 – Sep 2023
- Designed BERT-based NLU rescoring for ASR lattice decoding, **reducing WER by 1.36% over Whisper on LibriSpeech**; led 5-person international team from formulation through peer-reviewed publication.

## INDUSTRY EXPERIENCE

**Data Scientist** Nov 2020 – Jul 2022
*KPMG • Bengaluru, India*
- To reduce high-friction information lookup for 500+ employees, built and deployed a conversational AI system (GPT-2 fine-tuned + RAG, PyTorch) increasing productivity by 23%; won inaugural firmwide hackathon across 30 teams.
- Built neural entity-matching engine (TF-IDF + deep learning) with KPMG Europe, cutting manual account-mapping time by ~83% per audit cycle.

**Data Scientist** Jan – Oct 2020
*Graphene Health Tech • Bengaluru, India*
- Designed NLP classification pipeline (CNNs, GRUs, Transformers) for FMCG social media intelligence, filtering noisy data by 85% and improving processing efficiency by 25%; Jaccard+LSH deduplication improved pipeline throughput by 37%.

## TECHNICAL SKILLS & HONORS

**Languages:** Python (primary), C++, SQL | **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn
**Research Areas:** NLU, Computer Vision, Deep Learning, Multimodal AI, ASR, Music IR, 3D Human Motion Modeling, Representation Learning, Generative 3D, Motion Evaluation, Human Avatar Synthesis
**Tools:** Git, AWS, SLURM/HPC, Docker, W&B, PyTorch3D, Kaldi, MediaPipe, LaTeX | **Methods:** Transformers, LSTMs, VQ-VAE, Contrastive Learning, Audio-Motion Alignment, Synthetic Data Curation, Vision-Language Models
**Honors:** Best Poster, Purdue Symposium (2024) • MBAA Intl. Finalist (2024) • IBM Qiskit Winner (2023) • TA: MS NLP & Dialogue Systems, UB